

Multiple Choice (10 marks)

- Which of the following is incorrectly named?
 - MgO magnesium oxide
 - AgNO₃ silver nitrate
 - CuCl₂ copper(II) chloride
 - H₂SO₄ sulfuric acid
 - K₂Cr₂O₇ potassium dichromate
- Which of the following geometries corresponds to a substance that has five sigma bonds and one nonbonding pair of electrons?
 - Bent
 - Trigonal pyramid
 - T-shaped
 - Trigonal planar
 - Trigonal bipyramidal
- What is the mass of one molecule of cholesterol (C₂₇H₄₆O, molecular mass = 386)?
 - $3 \times 10^{-23}g$
 - $7.15 \times 10^{-22}g$
 - $1.38 \times 10^{-21}g$
 - $6.41 \times 10^{-22}g$
 - $8.41 \times 10^{-22}g$
- Solve $6x^2 - 7x - 20 = 0$.
 - $5/2, 4/3$
 - $-5/3, 4/2$
 - $5/3, -4/2$
 - 2, -10
 - 4, -5
- Calculate the pressure exerted by 1.0 molXe when it is confined to 1.0dm³ at 25°C.

- (a) 21 atm
- (b) 28 atm
- (c) 30 atm
- (d) 12 atm
- (e) 18 atm

True / False (10 marks)

Answer True or False

6. True or False: The resulting molarity of the sucrose solution after mixing is 0.33 M.
7. True or False: The nuclide mass of N^{13} is 13.00335 amu, reflecting its specific atomic mass post β^+ decay.
7. True or False: The pH of a 0.001 M aqueous solution of KOH is 13, indicating it is a strong base.
8. True or False: The work function of potassium metal is approximately 2.50×10^{-19} J, which is lower than its threshold energy.
8. True or False: The volume of O_2 required to completely burn 50 liters of CO is 75 liters, producing 150 liters of CO_2 .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of determining the pH of a 0.2 M NH_3 solution, given that $K_b = 1.8 \times 10^{-5}$ at 25°C. Include the equilibrium reaction and your calculations.
11. Discuss the reaction that occurs when Fehling's solution is added to diabetic urine containing sugar. Analyze the chemical processes involved and identify the precipitate formed upon heating.

End of Paper